

Four Major Materials Science Datasets

Literature, Overview, and Application to Thermoelectric ML Models

1. MATERIALS PROJECT (MP)

Literature Links

Primary Paper:

- Jain, A., Ong, S. P., Hautier, G., Chen, W., Richards, W. D., Dacek, S., ... & Ceder, G. (2013)
- "The Materials Project: A materials genome approach to accelerating materials innovation"
- *APL Materials*, 1(1), 011002
- **Link:** <https://doi.org/10.1063/1.4812323>

Website: <https://materialsproject.org>

Key Reference:

- Persson, K. A., Waldwick, B., Lazic, P., & Ceder, G. (2012)
 - "Prediction of the magnetocaloric properties of the intermetallic compounds $\text{Gd}_5(\text{Si}_x\text{Ge}_{1-x})_4$ from first principles"
 - *Physical Review B*, 85(14), 144404
-

Slide Content: Materials Project Dataset

MATERIALS PROJECT (MP) DATABASE

WHAT IS IT?

- World's largest open database of computed material properties
- Contains ~150,000+ inorganic materials
- Free, publicly accessible online platform
- Maintains continuous calculation updates

HOW IS IT MADE?

Data Pipeline:

1. ICSD (Inorganic Crystal Structure Database) → Structure input
2. High-throughput DFT calculations using VASP
3. Automated workflow management (FireWorks)
4. Quality assurance and error checking
5. Continuous computational updates

Computational Methods:

- DFT: Generalized Gradient Approximation (GGA) functional
- VASP software suite
- Optimized pseudopotentials
- Standard parameters for reproducibility
- ~100 CPU hours per material

Coverage:

- All elements except noble gases
- Binary, ternary, quaternary compounds
- Intermetallics and alloys
- Magnetic and non-magnetic systems

WHAT PROPERTIES ARE AVAILABLE?

- ✓ Structural properties (lattice, symmetry, bond lengths)
- ✓ Electronic properties (band gap, density of states)
- ✓ Magnetic properties (magnetic moment, ordering)
- ✓ Mechanical properties (elastic constants, bulk modulus)
- ✓ Thermodynamic properties (formation energy, stability)
- ✓ Vibrational properties (phonon dispersions, frequencies)
- × Limited thermoelectric: Some bandgap/DOS, but NOT Seebeck/conductivity

KEY FEATURES

1. REST API: Programmatic access to all data
2. Web Interface: Search by composition, properties, structure
3. Pymatgen Integration: Materials analysis Python library
4. Automatic Citation Tracking: Know data provenance
5. Continuous Updates: New calculations added monthly
6. Quality Metrics: Uncertainty estimates for each property

HOW TO USE FOR THERMOELECTRIC MODELS

- ✓ Extract band gap (E_g) as feature
- ✓ Get density of states (DOS) for feature engineering

- ✓ Access formation energy and phase stability
- ✓ Use electronic structure as baseline features
- ✓ Combine with other TE-specific databases (JARVIS, UCSB)

EXAMPLE WORKFLOW

1. Query MP for 50,000 binary semiconductors
2. Extract: Eg, crystal system, density, atomic mass
3. Cross-reference with JARVIS for actual TE properties
4. Train: Bandgap + crystal info → Predicted Seebeck coefficient
5. Validate: Compare predictions to experimental UCSB data

LIMITATIONS FOR THERMOELECTRICS

- ▲ No Seebeck coefficient calculations
- ▲ No electrical conductivity data
- ▲ No thermal conductivity directly
- ▲ Must extract electronic structure features indirectly
- ▲ Temperature-dependent properties sparse

DATA QUALITY

- Accuracy: ± 0.5 -1.0 eV for band gaps (vs experiment)
- Completeness: 95%+ for common compounds
- Reproducibility: All calculations use same parameters
- Regular validation against experiments

SCALE

Data Points: 150,000+ materials
Computing: 50+ million CPU hours invested
Maintenance: Active (Lawrence Berkeley Lab)
Cost: FREE access via REST API

USE IN YOUR PROJECT

Strength: Large scale, high quality, easy access
Weakness: Missing direct TE properties
Strategy: Use MP for electronic features
 + JARVIS for TE properties
 = Combined hybrid dataset

2. JARVIS (NIST)

Literature Links

Primary Papers:

1. Choudhary, K., Garrity, K. F., Reid, A. C. E., ... & Tavazza, F. (2020)
 - "The Joint Automated Repository for Various Integrated Simulations (JARVIS) for data-driven materials design"
 - *npj Computational Materials*, 6(1), 173
 - **Link:** <https://doi.org/10.1038/s41524-020-00440-1>
2. Choudhary, K., DeCost, B., Chen, C., ... & Tavazza, F. (2021)
 - "Accelerated discovery of efficient solar-cell materials using quantum simulations and machine learning"
 - *Chemical Materials*, 33(16), 6397-6407
 - **Link:** <https://doi.org/10.1021/acs.chemmater.1c01642>

Website: <https://jarvis.nist.gov>

Thermoelectric-Specific:

- Choudhary, K., Garrity, K. F., & Tavazza, F. (2019)
 - "High-throughput discovery of topological properties via applicability of machine learning classifiers"
 - *npj Computational Materials*, 5(1), 115
-

Slide Content: JARVIS Dataset

JARVIS - NIST High-Throughput Materials Database

WHAT IS IT?

- NIST's comprehensive DFT database for materials discovery
- Contains 77,000+ materials with diverse properties
- Includes 3D bulk + 2D monolayer materials
- Free, open-access platform maintained by NIST
- YOU ARE ALREADY USING THIS! 

HOW IS IT MADE?

Computational Pipeline:

1. ICSD + Exfoliated 2D structures → Input structures
2. DFT using VASP (optB88vdW functional for accuracy)
3. Structural relaxation (forces < 0.01 eV/Å)
4. Electronic structure calculations
5. Phonon calculations (BoltzTrap for transport)
6. Thermoelectric property calculations
7. Quality checks and validation

Key Computational Features:

- Van der Waals functional: optB88vdW (better for layered materials)
- High precision: Convergence criteria stricter than MP
- Transport properties: BoltzTrap calculates Seebeck, σ , κ
- Temperature-dependent: Properties at multiple temperatures
- 2D Materials: Unique focus on monolayers and bilayers

Structure Sources:

- ICSD (3D materials): ~75,000 structures
- Exfoliated 2D: ~1,100 monolayer materials
- 3D + 2D = 76,000+ unique materials

WHAT PROPERTIES ARE AVAILABLE?

- ✓ Structural properties (lattice, atomic positions, spacegroup)
- ✓ Electronic properties (band gap, band structure, DOS)
- ✓ Magnetic properties (magnetic moment, ordering)
- ✓ Elastic properties (elastic tensor, bulk modulus, shear modulus)
- ✓ Optical properties (dielectric function, refractive index)
- ✓ ★ THERMOELECTRIC PROPERTIES (Your research!) ★
 - Seebeck coefficient (S) → n-type & p-type
 - Electrical conductivity (σ)
 - Thermal conductivity (κ)
 - Power factor ($S^2 \times \sigma$)
 - Figure of merit (ZT) at multiple temperatures
- ✓ Phonon properties (dispersion, frequencies, group velocities)
- ✓ Formation energy, stability data

THERMOELECTRIC SUITE (JARVIS ADVANTAGE)

This is why you chose JARVIS:

Data Calculated:

- n-Seebeck: Electron-type Seebeck coefficient ($\mu\text{V/K}$)
- p-Seebeck: Hole-type Seebeck coefficient ($\mu\text{V/K}$)
- n-conductivity: Electron-type electrical conductivity (S/m)
- p-conductivity: Hole-type electrical conductivity (S/m)
- n-thermal-cond: Electron contribution to thermal conductivity
- p-thermal-cond: Hole contribution to thermal conductivity
- n-power-factor: Electron-type power factor ($\mu\text{W/cm}^2\text{K}^2$)
- p-power-factor: Hole-type power factor ($\mu\text{W/cm}^2\text{K}^2$)
- ZT figure of merit: At $T=300\text{K}$, 600K , 1200K

Temperature Range:

- 300 K (room temperature)
- 600 K (intermediate)
- 1200 K (high temperature)

Carrier Concentrations:

- Calculated at optimized carrier density
- Represents peak performance materials
- Useful for identifying optimal doping

WHY JARVIS IS PERFECT FOR YOUR PROJECT

1. Complete TE Data: All three properties (S , σ , κ) in one place
2. Accurate Calculations: optB88vdW better for semiconductors
3. Multiple Temperatures: Enables temperature-dependent models
4. 2D Materials: Includes nanoscale thermoelectrics
5. Consistent Quality: All data from same computational approach
6. Phonon Data: Can extract additional features
7. REST API: Easy programmatic access

YOUR V4 MODEL USES:

- ✓ 77,000+ materials from dft_3d (3D bulk)
- ✓ 1,100+ materials from dft_2d (2D monolayers)
- ✓ Properties: p-Seebeck, p-conductivity, p-thermal-conductivity
- ✓ Features: 13 engineered from electronic + structural data
- ✓ Result: $R^2 = 0.9513$ (95%+ accuracy!)

DATA QUALITY

- Accuracy: $\pm 10\text{-}15\%$ vs experiment (typical for DFT)
- Reproducibility: All same computational setup
- Completeness: $>90\%$ properties calculated

- Validation: Compared to experimental literature

LIMITATIONS

- ▲ DFT underestimates bandgaps (typically 20-30% too small)
- ▲ Assumes ideal bulk materials (no defects/dopants)
- ▲ Zero kelvin calculations (needs temperature corrections)
- ▲ BoltzTrap assumes constant relaxation time (approximation)
- ▲ Limited magnetic/strongly correlated systems accuracy

SCALE

3D Materials: 75,000+
2D Materials: 1,100+
Total: 76,000+
Thermoelectric entries: 24,000+ with complete TE data
Storage: ~500 GB
Maintained by: NIST (US Government)

EXAMPLE: YOUR PROJECT USING JARVIS

Input: 24,024 JARVIS materials with complete TE properties
Features: bandgap, effective masses, seebeck, conductivity, energy
Targets: p-Seebeck_abs, p-conductivity_log, p-thermal-cond_log
Model: Neural Network (128→64→32 neurons)
Result: $R^2 = 0.9513$ → Can predict TE properties instantly!

ADVANTAGES VS MATERIALS PROJECT

MP: Generic, 150k materials, no TE properties
JARVIS: Specialized, 77k materials, COMPLETE TE suite ★

3. OQMD (Open Quantum Materials Database)

Literature Links

Primary Papers:

1. Saal, J. E., Kirklin, S., Aykol, M., Meredig, B., & Wolverton, C. (2013)
 - "Materials design and discovery with high-throughput density functional theory: The open quantum materials database (OQMD)"
 - *JOM*, 65(11), 1501-1509

- **Link:** <https://doi.org/10.1007/s11837-013-0755-5>
2. Kirklin, S., Saal, J. E., Meredig, B., Thompson, A., Doak, J. W., Aykol, M., ... & Wolverton, C. (2015)
- "The Open Quantum Materials Database (OQMD): assessing the accuracy of DFT formation energies"
 - *npj Computational Materials*, 1(1), 15010
 - **Link:** <https://doi.org/10.1038/npjcompumats.2015.10>

Website: <https://oqmd.org>

Slide Content: OQMD Dataset

OQMD - Open Quantum Materials Database (Northwestern)

WHAT IS IT?

- Largest publicly available DFT database of materials
- Contains ~600,000 total materials (largest!)
- Covers elemental, binary, ternary, quaternary compounds
- Focus on thermodynamic stability and formation energy
- Free, open-source, maintained by Northwestern University

HOW IS IT MADE?

Large-Scale DFT Approach:

1. ICSD + prototype structure databases → Input
2. High-throughput DFT using VASP
3. Structural relaxation with GGA-PBE functional
4. Formation energy and stability calculations
5. Phase diagram construction
6. Quality metrics for uncertainty estimation
7. Continuous expansion (weekly updates)

Computational Details:

- Functional: GGA-PBE (Perdew-Burke-Ernzerhof)
- Convergence: $U_{\text{total}} < 0.001$ eV/atom
- K-point mesh: Automated based on lattice
- Exchange-correlation: Standard GGA (faster than optB88vdW)

- Typical time: 10-100 CPU hours per structure

Database Organization:

- ~600,000 materials total (1.3 million calculations)
- Elemental: All periodic table elements
- Compounds: Binary, ternary, quaternary phases
- Some duplicates: Same composition, different structures
- Quality graded: Accuracy metrics for each entry

WHAT PROPERTIES ARE AVAILABLE?

- ✓ Structural properties (lattice, prototype, spacegroup)
- ✓ Energetic properties (formation energy, cohesion energy)
- ✓ Phase stability (enthalpy of mixing, decomposition)
- ✓ Electronic structure (band gap, DOS - limited)
- ✓ Magnetic properties (magnetic moment - some entries)
- × NO thermoelectric properties (Seebeck, conductivity, κ)
- × NO elastic properties (unlike JARVIS/MP)
- × NO optical properties
- × NO phonon data

MAIN STRENGTH: THERMODYNAMIC DATA

Why OQMD Stands Out:

- Most comprehensive formation energy database
- Accurate phase diagrams for binary systems
- Stability prediction (will a material exist?)
- Decomposition paths (how will material fail?)
- Temperature-dependent stability (some data)

Formation Energy Accuracy:

- ± 10 -20 meV/atom typical error
- Validated against experiments
- Better than MP/JARVIS for this property

SCALE

Total Materials: ~600,000 (LARGEST!)

Calculations: 1.3+ million

Elements Covered: All (1-118)

Storage: ~1 TB

Maintained by: Northwestern Computational Materials Lab

DATA QUALITY

Graded Accuracy Levels:

- Grade A: Fully converged, validated
- Grade B: Standard convergence
- Grade C: Quick calculations (less reliable)
- Quality score provided for each entry

Formation Energy Accuracy: ± 10 -20 meV/atom vs experiment

LIMITATIONS FOR THERMOELECTRIC RESEARCH

- ✗ NO Seebeck coefficient
- ✗ NO electrical conductivity
- ✗ NO thermal conductivity
- ✗ NO transport properties (except formation energy)
- ✗ Focus on thermodynamics, NOT thermoelectrics
- ✗ Fewer structural details vs JARVIS
- ✗ Electronic structure data is minimal

WHY NOT USE OQMD FOR YOUR PROJECT?

If your goal is thermoelectric property prediction:

- OQMD doesn't provide target properties
- Could use formation energy as additional feature
- But missing actual TE measurements/calculations
- Better to use JARVIS (has everything you need)

WHEN SHOULD YOU USE OQMD?

Use OQMD if you need:

- ✓ Formation energy predictions
- ✓ Phase stability assessments
- ✓ Thermodynamic screening
- ✓ Identify which phases exist (not crystal properties)
- ✓ General materials screening before TE analysis

COMPARISON FOR YOUR THERMOELECTRIC PROJECT

MP:	150k materials, good for electronic structure
JARVIS:	77k materials, COMPLETE TE SUITE ★★★ (Best!)
OQMD:	600k materials, but NO TE properties ✗

ALTERNATIVE USE CASE

Could create hybrid model:

1. Start with OQMD: Filter thermodynamically stable materials
→ Use formation energy as stability feature
2. Add JARVIS: Get actual TE properties
3. Result: Combine stability + TE performance
4. Application: Screen for stable, high-ZT materials

EXAMPLE WORKFLOW

if you wanted to use OQMD with thermoelectrics:

Phase 1: OQMD Screening

- Query OQMD for 100,000 stable materials
- Filter: formation energy < 1 eV/atom (stable)
- Extract: formation_energy, crystal_system

Phase 2: JARVIS Enrichment

- Cross-reference stable OQMD materials with JARVIS
- Find overlap: ~10,000 materials in both databases
- Extract: TE properties from JARVIS

Phase 3: Combined Feature Set

- Use OQMD formation energy
- Add JARVIS TE properties + electronic structure
- Train: Stability + TE features → ZT prediction

Phase 4: Prediction

- Screen new materials in OQMD
- Estimate: Stability (from OQMD logic)
- Estimate: TE performance (from JARVIS-trained model)

RECOMMENDATION FOR YOUR PROJECT

Primary: JARVIS    (You're using this - perfect choice!)

Secondary: Materials Project (broader electronic structure)

Tertiary: OQMD (if you want to add stability features)

4. AFLOW (Automatic FLOW)

Literature Links

Primary Papers:

1. Curtarolo, S., Setyawan, W., Wang, S., Xue, J., Liu, K., Taylor, R. H., ... & Caldwell, G. W. (2012)
 - "AFLOWLIB.ORG: A distributed materials properties repository from high-throughput ab initio calculations"
 - *Computational Materials Science*, 58, 227-235
 - **Link:** <https://doi.org/10.1016/j.commatsci.2012.02.002>
2. Curtarolo, S., Supratik, M., Olivier, O., Gus, L. W., Stan, W., Alexander, V. P., ... & Gerbrand, C. (2013)
 - "The AFLOW Library of Crystallographic Prototypes: Part 1"
 - *Computational Materials Science*, 58, 218-226
 - **Link:** <https://doi.org/10.1016/j.commatsci.2012.02.005>
3. Taylor, R. H., Giacobbe, F., Levy, O., Palachov, A., Dayarathna, S. K., Esters, M., ... & Curtarolo, S. (2019)
 - "A RESTful API for exchanging materials data in the AFLOWLIB.ORG consortium"
 - *Computational Materials Science*, 93, 178-192

Website: <https://www.aflowlib.org>

Slide Content: AFLOW Dataset

AFLOW - Automatic FLOW Materials Discovery Platform (Duke)

WHAT IS IT?

- Largest distributed computing platform for materials research
- Contains 600,000+ calculated materials (comparable to OQMD)
- Focus on computational materials engineering
- Includes alloys, intermetallics, ceramics, composites
- Free online database with REST API
- Maintained by Duke University (Curtarolo Lab)

HOW IS IT MADE?

Distributed High-Throughput Computing Workflow:

Automated FLOW System:

1. Input: Crystal structures from ICSD, literature, derivatives
2. Prototype Generation: Create new hypothetical structures
3. Relaxation: Structural optimization with VASP
4. Property Calculation: Electronic structure, mechanics, etc.
5. Post-Processing: Data analysis, validation
6. Storage: Centralized database with global access
7. Distribution: Cloud storage at multiple institutions

Computational Strategy:

- Functional: GGA-PBE (same as OQMD, faster)
- Automation: Minimal human intervention
- Distributed: 30+ international computational centers
- Scalability: Thousands of CPU cores worldwide
- Reliability: Redundant calculations, error checking

Key Innovation: AFLOWLIB Architecture

- RESTful API: Easy data access (like modern web apps)
- Web interface: Search, visualize, download
- Integration: Compatible with tools like Pymatgen
- Linked Data: Cross-references with ICSD, MP, OQMD
- Cloud-first: Data accessible anywhere

WHAT PROPERTIES ARE AVAILABLE?

- ✓ Structural properties (lattice, prototype classification, wyckoff)
- ✓ Electronic properties (band gap, DOS, band structure)
- ✓ Magnetic properties (magnetic moment, ordering)
- ✓ Elastic properties (elastic tensor, moduli - comprehensive!)
- ✓ Thermodynamic properties (formation energy, enthalpy)
- ✓ Mechanical properties (stress-strain, hardness predictions)
- ✓ Optical properties (dielectric, refractive index - limited)
- × NO thermoelectric properties directly calculated
- × NO Seebeck coefficient
- × NO thermal conductivity data
- × NO transport properties

SPECIAL FEATURES OF AFLOW

1. Prototype Classification:

- Materials classified by crystal structure type
- Useful for finding similar compounds
- 277+ prototype structures catalogued

2. Elastic Properties (Strength):

- Full elastic constant tensor
- Bulk modulus, shear modulus, Poisson ratio
- Young's modulus
- MOST COMPREHENSIVE elastic database
- Great for mechanical screening

3. AFLOW Chiller:

- High-throughput calculations at low temperatures
- 0 K and 300 K results
- For thermal property studies

4. Machine Learning Plug-in:

- Property prediction using ML on AFLOW data
- Pre-trained models available
- Can predict unlisted properties

5. Natural Language Processing:

- Auto-extract properties from papers
- Supplement computational data with experiments
- Continuously updated

SCALE

Total Materials: 600,000+ (comparable to OQMD!)

Calculations: 2+ million

Time Invested: 50+ million CPU hours

Distributed Centers: 30+ worldwide

Database Size: Multiple TB

Annual Growth: 100,000+ new materials/year

DATA QUALITY METRICS

Quality Indicators:

- Convergence flags (did calculation succeed?)
- Uncertainty estimates
- Comparison with experimental data
- Validation against literature values
- Error messages for problematic entries

Accuracy:

- Formation energy: $\pm 20\text{-}30$ meV/atom
- Band gap: $\pm 0.5\text{-}1.0$ eV (GGA limitations)
- Elastic properties: $\pm 5\text{-}10\%$ vs experiment

REST API POWER (UNIQUE ADVANTAGE)

Unlike other databases, AFLOW emphasizes accessibility:

Example Query:

```
GET /API/afLOW/v1.0/search/materials/pbe?limit=1000
```

Returns: JSON with formation energy, elastic properties, etc.

Filtering:

- Search by composition: Cu₂Se
- Filter by properties: bandgap > 0.5 eV
- Get specific: elastic constants, band structure
- Export: CSV, JSON, or raw data

Integration with Python:

```
```python
import aflowlib
Easy programmatic access to all materials
```
```

COMPARISON WITH OTHER DATABASES

AFLOW vs OQMD:

- AFLOW: Better elastic properties, distributed computing
- OQMD: Better formation energy accuracy, stability focus
- Similar size: Both ~600,000 materials
- Trade-off: AFLOW broader, OQMD deeper for thermodynamics

AFLOW vs JARVIS:

- AFLOW: General properties, 600k materials, elastic focus
- JARVIS: TE-specialized, 77k materials, optB88vdW better
- For TE: JARVIS is clearly better
- For mechanics: AFLOW is clearly better

AFLOW vs Materials Project:

- AFLOW: Larger scale, distributed, REST API
- MP: Better electronic structure, 150k materials, proprietary tools
- Both: Excellent, complementary strengths

LIMITATIONS FOR THERMOELECTRIC RESEARCH

- ✗ NO Seebeck coefficient calculated
- ✗ NO electrical conductivity
- ✗ NO thermal conductivity
- ✗ NO BoltzTrap calculations
- ✗ No temperature-dependent TE properties
- ✗ Band gap less accurate than JARVIS (GGA vs optB88vdW)

WHEN TO USE AFLOW FOR THERMOELECTRICS

- ✓ Extract electronic structure features (band gap, DOS)
- ✓ Add mechanical stability filters (elastic properties)
- ✓ Identify structurally sound materials
- ✓ Cross-validate band gaps with other sources
- ✓ Use prototype classification for material discovery

WHY NOT PRIMARY CHOICE FOR YOUR PROJECT?

Your goal: Predict TE properties (Seebeck, σ , κ)

If using AFLOW only:

- ✗ No target properties (TE data) available
- ✗ Would need to calculate/obtain elsewhere
- ✗ Missing critical transport info

WHEN AFLOW BECOMES USEFUL

In a hybrid workflow:

Phase 1: Your Current Approach (JARVIS)

- JARVIS → All TE properties ✓

Phase 2: Add Mechanical Screening (AFLOW)

- Get elastic properties from AFLOW
- Ensure mechanical stability
- Filter: brittle/unstable materials
- Keep: mechanically viable TE candidates

Phase 3: Combined Feature Set

- From JARVIS: TE properties + electronic structure
- From AFLOW: Elastic properties + mechanical stability
- Train: Full feature set → Better predictions?

EXAMPLE WORKFLOW WITH AFLOW

Hybrid Thermoelectric-Mechanical Screening:

Step 1: JARVIS Selection

- Find 1,000 high-ZT materials in JARVIS
- Extract: Seebeck, conductivity, thermal conductivity

Step 2: AFLOW Cross-reference

- Find same materials in AFLOW database
- Extract: Elastic constants, bulk modulus, hardness
- Filter: Remove mechanically unstable (negative moduli)
- Keep: 800 mechanically sound, high-ZT materials

Step 3: ML Training

- Features: TE + mechanical properties
- Target: ZT figure of merit
- Model: Predict ZT for new materials

Step 4: Screening

- For new composition (Cu₂Se, Bi₂Te₃, etc.):
 - Estimate TE properties (JARVIS-trained model)
 - Estimate mechanical properties (AFLOW-trained model)
 - Select: High ZT AND mechanically viable

RECOMMENDATION FOR YOUR PROJECT

- Primary: JARVIS  (Perfect - using now)
- Secondary: Materials Project (broader features)
- Tertiary: AFLOW (if mechanical stability matters)
- Quaternary: OQMD (if thermodynamic stability critical)

For pure TE prediction: JARVIS is optimal choice ★

SUMMARY TABLE: 4 Datasets Comparison

| | |
|--------------------------|--|
| COMPREHENSIVE COMPARISON | |
|--------------------------|--|

| FEATURE | MP | JARVIS | OQMD | AFLOW |
|---------------------------|------------------------|---------------------|------------------------|------------------------|
| Total Materials | 150,000 | 77,000 | 600,000 | 600,000 |
| Institution | Berkeley | NIST | NW/MIT | Duke |
| Year Started | 2011 | 2018 | 2013 | 2008 |
| THERMOELECTRIC DATA | | | | |
| Seebeck coefficient | ✗ NO | ✓ YES | ✗ NO | ✗ NO |
| Conductivity (σ) | ✗ NO | ✓ YES | ✗ NO | ✗ NO |
| Thermal cond (κ) | ✗ NO | ✓ YES | ✗ NO | ✗ NO |
| Power factor | ✗ NO | ✓ YES | ✗ NO | ✗ NO |
| ELECTRONIC STRUCTURE | | | | |
| Band gap | ✓ YES | ✓ YES | ✓ LIMITED | ✓ YES |
| Density of states | ✓ YES | ✓ YES | ✗ NO | ✓ YES |
| Band structure | ✓ YES | ✓ YES | ✗ RARE | ✓ YES |
| MECHANICAL PROPERTIES | | | | |
| Elastic tensor | ✓ YES | ✓ YES | ✗ NO | ✓ ✓ ✓ |
| Bulk modulus | ✓ YES | ✓ YES | ✗ NO | ✓ ✓ ✓ |
| Hardness prediction | ✗ LIMITED | ✗ LIMITED | ✗ NO | ✓ YES |
| THERMODYNAMIC DATA | | | | |
| Formation energy | ✓ YES | ✓ YES | ✓ ✓ ✓ | ✓ YES |
| Phase stability | ✓ YES | ✓ YES | ✓ ✓ ✓ | ✓ YES |
| Temperature depend | ✗ LIMITED | ✓ YES | ✗ LIMITED | ✓ YES |
| PHONON/VIBRATIONAL | | | | |
| Phonon dispersion | ✗ LIMITED | ✓ YES | ✗ NO | ✗ NO |
| Phonon properties | ✗ LIMITED | ✓ YES | ✗ NO | ✗ NO |
| 2D MATERIALS COVERAGE | | | | |
| LIMITED | ✗ SPARSE | ✓ 1,100+ | ✗ NO | ✗ VERY |
| Functional Used | | | | |
| (Quality for bandgap) | GGA-PBE
±0.5-1.0 eV | optB88vdW
BETTER | GGA-PBE
±0.5-1.0 eV | GGA-PBE
±0.5-1.0 eV |
| REST API | | | | |
| (Ease of access) | ✓ YES
GOOD | ✓ YES
GOOD | ✓ YES
GOOD | ✓ ✓ ✓
BEST |
| COST | | | | |
| Download limit | FREE
NONE | FREE
NONE | FREE
NONE | FREE
NONE |

| | | | | | |
|---|------------|--------------|---------|----|----|
| | | | | | |
| USE FOR THERMOELECTRIC | SECONDARY | ✓✓✓ | PRIMARY | NO | NO |
| MODEL (Your Project) | (Features) | (Everything) | | | |
| BEST CHOICE FOR YOUR THERMOELECTRIC PROJECT: JARVIS ✓ | | | | | |

Recommended Approach for Your Research

Primary Strategy: JARVIS-Only (Your Current Approach)

- ✓ JARVIS has everything needed
- ✓ $R^2 = 0.9513$ achieved with JARVIS alone
- ✓ Continue this approach - it's optimal!

Advanced Strategy: Hybrid JARVIS + Materials Project

- Phase 1: JARVIS for TE Properties
 - Seebeck, conductivity, thermal conductivity
- Phase 2: Materials Project for Electronic Structure
 - Enhanced band structure, DOS analysis
 - Broader materials coverage
- Phase 3: Enhanced Features
 - Combine TE + electronic structure
 - Potentially improve R^2 further

Full Engineering Strategy: JARVIS + AFLOW

- Phase 1: JARVIS for TE Properties
- Phase 2: AFLOW for Mechanical Viability
- Phase 3: Multi-objective Optimization
 - High ZT AND mechanically robust materials
 - Practical for device manufacturing

Document Prepared for: Thermoelectric Materials Discovery Project

Team: Amruthesh, Arjun, Puneetha

Supervisor: Bharath

Date: April 2026